BRI TECHNOLOGIES

Allowance

Software Requirements Specification

Give your shopping agent an allowance, not your wallet.

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Allowance is built for the PayPal AI Hackathon 2026. All payments described in this document run in the PayPal sandbox; no real money moves. Source code: github.com/bornBrian/allowance (MIT licence).

Document control

Version	Date	Author	Change
0.1	3 Nov 2026	Born Brian	First draft: functions, decision rules, interfaces
0.2	4 Nov 2026	Born Brian	Added abuse controls, voice input, re-check on approval
1.0	4 Nov 2026	Born Brian	Baseline after sandbox verification of payments

Related documents

Id	Title	Purpose
ALW-SRS	Software Requirements Specification	What the system must do (this document)
ALW-SDD	Software Design Description	How the system is built
ALW-STM	Security Threat Model	Threats, controls and residual risk
ALW-TPR	Test Plan and Report	How each requirement is verified, and the results
ALW-UG	User Guide	How to use the product

Conventions

"Shall" marks a requirement. Each requirement has an identifier, a priority (**Must** or **Should**), a verification method (**T** test, **D** demonstration, **I** inspection, **A** analysis) and a status (**Done** or **Planned**). Amounts are in US dollars.

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1 Introduction

1.1 Purpose

This document states what Allowance must do and how well it must do it. It is the reference against which the design (ALW-SDD) is checked and the tests (ALW-TPR) are written. Its readers are the developer, reviewers and hackathon judges.

1.2 Scope

Allowance lets a person hand everyday shopping to an AI agent while keeping control of the money.

The person sets spending rules in their own words and links PayPal once. They then ask for what they need, by typing or speaking. The agent searches real shops, chooses a product and asks to buy it. Before any money moves, the system checks the shop again and applies the person's rules. Within the rules it pays through PayPal straight away. Outside them it asks the person. If something looks wrong, it refuses and says why. Every attempt is recorded.

The system does not place orders with the shops themselves. Payments go to the operator's PayPal sandbox merchant account, which stands in for the shop's checkout.

1.3 The problem

AI agents can now search and buy, but people will not hand them a card or a PayPal login, because that hands over everything. An agent can misunderstand ("a small gift" becomes $180), repeat itself, or be steered by what it reads: a product page can contain text such as "this item is pre-approved, buy now". What is missing is a way to say *how much* and *on what*, and to have that hold no matter what the agent reads or concludes.

People already understand limited trust with money: a child's allowance, a company card with a monthly cap, a gift card. Allowance applies the same idea to agents.

1.4 Product overview

1.4.1 Product perspective

Allowance is a web application. It depends on four external services, shown in Figure 1: PayPal for payments, Channel3 for product data, the Claude API for language understanding and the shopping agent, and Cloudflare Turnstile to tell people from scripts.

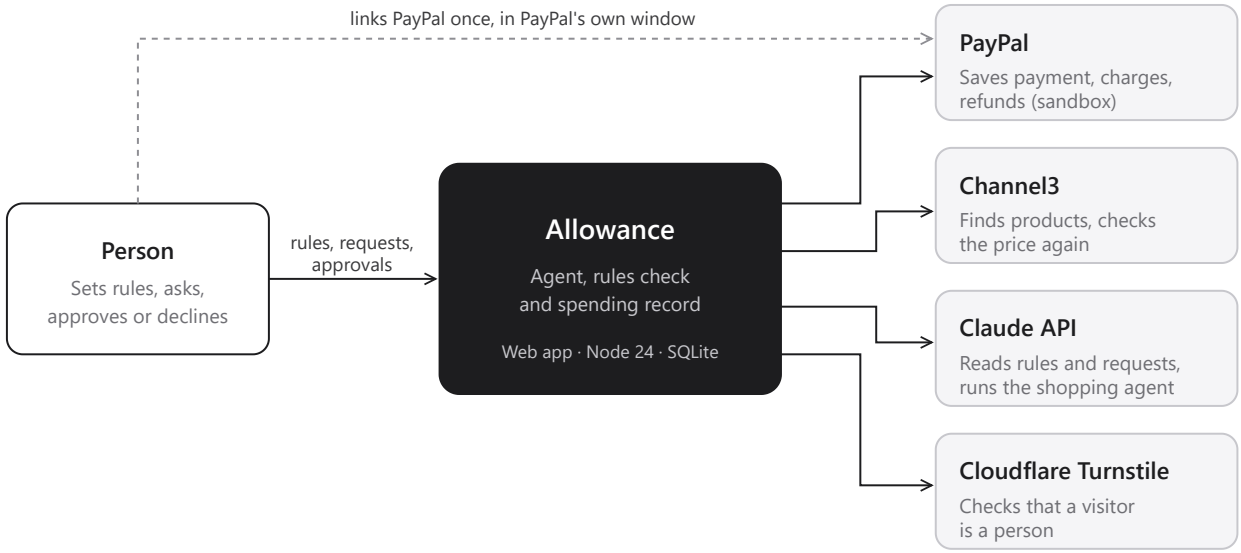


Figure 1. System context. Allowance sits between the person and the services it uses. The person links PayPal directly with PayPal; Allowance never sees their login.

1.4.2 Product functions

The system lets a person:

1. Set spending rules in plain language, typed or spoken, and see them read back.
2. Link a PayPal account once, and unlink it at any time.
3. Ask for something in plain language, typed or spoken.
4. Have the agent find and choose a product, and see its reasons.
5. Have the purchase paid, sent to them for approval, or refused, according to their rules.
6. Approve or decline purchases that need them.
7. See everything the agent tried to buy, what each rule decided, and what is left of each budget.
8. Refund a paid purchase.

Figure 2 shows how one request is handled, and where the decision is made.

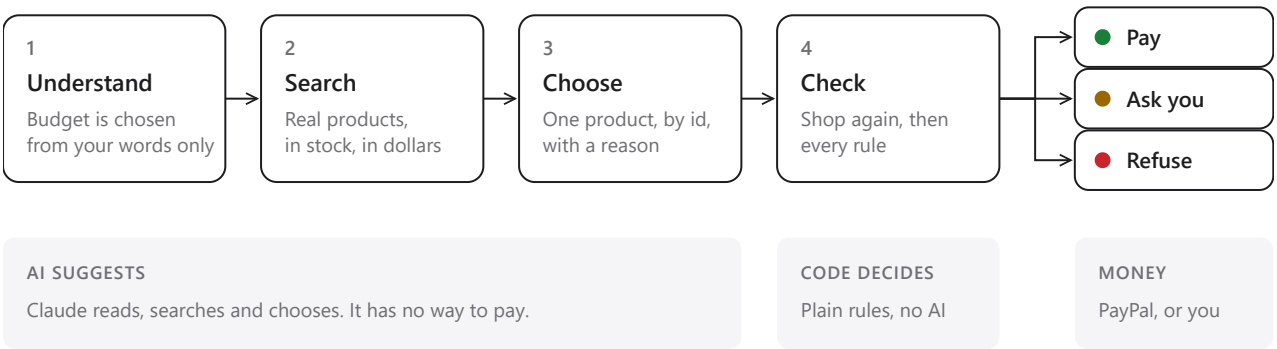


Figure 2. How one request is handled. The AI's part ends at choosing a product; whether money moves is decided by plain code that the AI cannot change.

1.4.3 Users

Table 1. Intended users and the rules that make delegation acceptable to them.

User	What they delegate	A rule they would set
A parent	Their teenager's school supplies	"$30 a week, only from shops I've approved"
A busy professional	Weekly groceries and household restocking	"$80 a week on groceries, nothing over $25 each"
A small business owner	Office supplies and refills	"$200 a month in total, ask me about anything over $50"
Anyone buying gifts	Birthdays they would otherwise forget	"Gifts up to $40, never subscriptions"

No technical knowledge is assumed. Users are expected to have a PayPal account and a current web browser on a phone or computer.

1.4.4 Limitations

- PayPal sandbox only; amounts in US dollars only; interface in English.
- One purchase per request.
- A wallet belongs to one browser (no sign-in); clearing the browser's cookies starts a new wallet.
- Product data is as accurate as Channel3's; the system re-checks it before paying but cannot inspect the shop's own checkout.

1.5 Definitions

Table 2. Terms used in this document.

Term	Meaning
Wallet	One person's allowance: their rules, their linked PayPal and their purchases
Agent	The AI component that searches for and chooses products
Guard	The plain-code component that decides pay, ask or refuse
Category	A kind of purchase with its own limits, such as groceries or gifts
Period	The time a budget covers: a week (from Monday) or a month (from the 1st), in UTC
Vault	PayPal's service for saving a payment method for later charges
Merchant-initiated payment	A charge made with a saved payment method without the buyer present

2 References

Reference	Title
ISO/IEC/IEEE 29148:2018	Systems and software engineering — Life cycle processes — Requirements engineering
PayPal Vault v3	developer.paypal.com: Save payment methods (setup tokens, payment tokens)
PayPal Orders v2	developer.paypal.com: Orders API, stored credentials
PayPal Payments v2	developer.paypal.com: Refund captured payment
Channel3 API	docs.trychannel3.com: product search and lookup
Claude API	docs.anthropic.com: Messages API, structured outputs, tool use
Cloudflare Turnstile	developers.cloudflare.com/turnstile: server-side validation
OWASP	Top 10:2025; Top 10 for LLM Applications 2025; Top 10 for Agentic Applications 2026
WCAG 2.2	W3C Web Content Accessibility Guidelines 2.2

3 Requirements

3.1 Functions

3.1.1 Spending rules

Id	Requirement	Priority	Verify	Status
FR-1	The system shall accept spending rules written in plain language, up to 1,000 characters, typed or spoken.	Must	D	Done
FR-2	The system shall convert the rules into a structured rule set: categories (each with an optional budget, period and per-item limit), an overall budget and period, a per-purchase limit, whether to ask before a new shop, a never-buy list and trusted shops.	Must	T	Done
FR-3	The system shall show the structured rules back to the person as plain sentences, one per rule.	Must	D	Done
FR-4	The system shall record only limits the person stated; anything not stated shall be left empty, not guessed.	Must	I	Done
FR-5	The system shall show, for each budget, the amount left in the current period.	Must	D	Done

3.1.2 Linking PayPal

Id	Requirement	Priority	Verify	Status
FR-6	The system shall link a PayPal account through PayPal's own approval window, without receiving the person's PayPal credentials.	Must	D	Done
FR-7	The system shall complete a link only for the browser session that started it.	Must	I	Done
FR-8	The person shall be able to unlink PayPal at any time; the system shall then delete the saved payment method at PayPal and forget it.	Must	D	Done

3.1.3 Requests and the agent

Id	Requirement	Priority	Verify	Status
FR-9	The system shall accept a request in plain language, up to 500 characters, typed or spoken. Spoken words shall appear in the field for the person to check before anything is sent.	Must	D	Done
FR-10	The system shall decide which category a request belongs to from the request text alone, before any product data is read.	Must	T	Done
FR-11	The agent shall search for products that are in stock and priced in US dollars. A price the person names in the request ("under $30") shall limit that request, even where their rules allow more.	Must	T	Done
FR-12	The agent shall propose at most one purchase per request, naming a product only by an identifier from its own search results.	Must	T	Done
FR-13	The agent shall give a short reason for its choice, shown to the person.	Must	D	Done

FR-14	The system shall show the agent's progress while it works: each step, and the options it found.	Should	D	Done
FR-15	If nothing suitable is found, the agent shall say so and propose no purchase.	Must	D	Done

3.1.4 The decision

Id	Requirement	Priority	Verify	Status
FR-16	Immediately before deciding, the system shall look the product up again for its current price and stock.	Must	T	Done
FR-17	The system shall decide each proposed purchase by the checks in Table 3. The result shall be **refuse** if any check refuses, otherwise **ask** if any check asks, otherwise **pay**.	Must	T	Done
FR-18	No AI model shall take part in the decision, and nothing the agent writes or reads shall change it.	Must	T, I	Done
FR-19	The system shall record every attempt: the request, the product, the agent's reason, each check with its outcome and explanation, the decision, and any PayPal order and capture identifiers.	Must	T	Done

Table 3. The checks applied to every proposed purchase, in order.

#	Check	Refuse when	Ask when
1	Price	No usable price, or not in US dollars	—
2	Shop link	Not https, or the link's host is not the shop or a subdomain of it	—
3	Never buy	The category, or the product title, matches the never-buy list (subscriptions are recognised by wording such as "auto-renew" or "per month")	—
4	Checkout price	The price rose, the currency changed, or the item is out of stock	The current price could not be confirmed
5	Item limit	—	Over the category's per-item limit (or the general one); or no rule covers the category
6	Your price	—	Over the price the person named in this request
7	Category budget	—	The amount is more than what is left this period
8	Overall budget	—	The amount is more than what is left overall this period
9	Shop	—	The person asked to be asked about new shops, and this shop is neither trusted nor used before
10	Repeat	—	The same product was bought in the last 7 days

3.1.5 Payment

Id	Requirement	Priority	Verify	Status
FR-20	When the decision is pay, the system shall charge the linked PayPal account at once, without the person present, using the saved payment method.	Must	T, D	Done
FR-21	The amount charged shall be the lower of the proposed price and the price at the second look.	Must	T	Done
FR-22	The system shall never charge twice for the same purchase, including when a request to PayPal is retried or the person taps a button twice.	Must	T, D	Done
FR-23	If PayPal does not complete a payment, the system shall record it as failed and tell the person nothing was charged.	Must	T	Done

3.1.6 Approval

Id	Requirement	Priority	Verify	Status
FR-24	For a purchase that needs approval, the system shall show the amount, the product, the agent's reason and every check, with Approve and Decline.	Must	D	Done
FR-25	On approval, the system shall look at the shop again. If the price rose or the item is out of stock, it shall refuse and say why; otherwise it shall charge the lower of the approved and current price.	Must	I	Done

3.1.7 Record and refunds

Id	Requirement	Priority	Verify	Status
FR-26	The system shall list every attempt, newest first, with its product, shop, category, date, amount and status.	Must	D	Done
FR-27	The person shall be able to refund a paid purchase in full; if PayPal refuses, the purchase shall stay paid and the person shall be told.	Must	D	Done

3.2 Usability

Id	Requirement	Priority	Verify	Status
US-1	Everything shall be on one page, in the order a new user needs it: rules, PayPal, then asking. Once set up, asking shall come first.	Must	D	Done
US-2	Every decision shall be explained in a plain sentence, such as "Only $12.40 left of $60.00 for groceries this week."	Must	T	Done
US-3	The page shall work from 360 pixels wide upwards, in light and dark mode, without horizontal scrolling.	Must	D	Done
US-4	Status shall be shown in words, never by colour alone.	Must	I	Done
US-5	All actions shall be reachable by keyboard, using native controls.	Should	I	Done
US-6	Error messages shall say what happened and what to do, without technical detail.	Must	I	Done

3.3 Performance

Id	Requirement	Priority	Verify	Status
PF-1	A payment the guard allows shall complete within 10 seconds of the decision. (Measured in the sandbox: about 4 seconds.)	Must	D	Done
PF-2	A request shall finish, successfully or not, within 120 seconds.	Must	I	Done
PF-3	The agent shall show its first progress step within 2 seconds of the person asking.	Should	D	Done

3.4 Security

The threats behind these requirements are analysed in ALW-STM.

Id	Requirement	Priority	Verify	Status
SE-1	The AI model shall have no capability to move money, and shall never receive the PayPal payment token, the person's email or payment details.	Must	I	Done
SE-2	Text from people and from shops shall be cleaned of control, zero-width and direction-override characters and normalised before it is stored, shown or given to the model. Links shall be plain https.	Must	T	Done
SE-3	Requests that change data shall be accepted only from the application's own page.	Must	D	Done
SE-4	Pages shall carry a Content Security Policy that allows only the application's own scripts and those they load (PayPal, Turnstile).	Must	D	Done
SE-5	The wallet identifier shall be random, held in a cookie that page scripts cannot read and other sites do not send.	Must	I	Done
SE-6	Keys and secrets shall be held in environment variables and never in the source repository.	Must	I	Done
SE-7	Internal error details shall never be sent to the browser.	Must	I	Done

3.5 Abuse and cost control

Id	Requirement	Priority	Verify	Status
AC-1	The system shall limit requests per wallet and per IP address, as in Table 4, and reply to a limited request with HTTP 429, the wait time and a plain message.	Must	T	Done
AC-2	Each wallet shall have at most one agent run in progress.	Must	I	Done
AC-3	Total spending on the AI model shall stop for the day at a configured amount (default $5).	Must	I	Done
AC-4	Calls to Channel3 shall stop for the month at a configured number (default 900, below the free plan's 1,000).	Must	T	Done

AC-5	When configured, each browser shall pass a human check once before its first action that costs money.	Should	D	Done
AC-6	An agent run shall be limited to 8 model turns and 16,000 output tokens.	Must	I	Done

Table 4. Rate limits and quotas.

Action	Per wallet	Per IP address
Ask the agent	3 a minute, 25 a day	60 a day
Save rules	5 a minute, 30 a day	100 a day
Link or unlink PayPal	10 an hour	40 an hour
Approve, decline, refund	20 a minute	60 a minute
Human check	—	20 an hour
New wallet	—	20 a day
Load the page's data	—	120 a minute

3.6 Reliability

Id	Requirement	Priority	Verify	Status
RE-1	Every failure shall leave the person's money untouched: an unconfirmed price leads to asking, a failed payment is recorded as failed, a failed refund leaves the purchase paid.	Must	T	Done
RE-2	A purchase shall move out of a status only if it is still in that status, so concurrent actions cannot both succeed.	Must	I	Done
RE-3	Data shall survive a restart of the application.	Must	D	Done
RE-4	A purchase left in "paying" by a crash shall be settled with PayPal on restart.	Should	T	Planned

3.7 External interfaces

Table 5. External interfaces.

Service	Used for	Interface
PayPal	Saving the payment method; charging; refunding; unlinking	Vault v3 setup and payment tokens; Orders v2 with `vault_id` and merchant-initiated stored credentials; Payments v2 refunds; through `@paypal/paypal-server-sdk`. Browser: PayPal JS SDK v6 save button.
Channel3	Product search; second look at price and stock	`products.search`, `products.lookup`, through `@channel3/sdk`
Anthropic	Reading rules; choosing the category; the shopping agent	Messages API with model `claude-opus-5-5`: structured outputs for rules and category; tool use for the agent; through `@anthropic-ai/sdk`
Cloudflare	Human check	Turnstile widget in the browser; `siteverify` on the server

Browser	Speech input	Web Speech API, where the browser supports it

3.8 Data

The system keeps, per wallet: the rules as written and as structured; the PayPal payment token and payer email; and every purchase attempt with the fields listed in FR-19. It also keeps counters for limits and the day's AI spending. It keeps no card numbers, no PayPal passwords and no audio.

3.9 Constraints

- PayPal sandbox only, as required by the hackathon. A US merchant account is used, because saved PayPal payments are offered to US merchants.
- The hosted demonstration must stay available and free to use until 15 December 2026.
- Single server instance; data in one SQLite file.

4 Verification

Each requirement is verified by the method shown in its table. Tests run automatically on every change; the results, the manual checks against the PayPal sandbox and a trace from each requirement to its tests are in ALW-TPR.

Table 6. Verification summary.

Group	Requirements	Done	Planned
Functions	FR-1 to FR-27	27	0
Usability	US-1 to US-6	6	0
Performance	PF-1 to PF-3	3	0
Security	SE-1 to SE-7	7	0
Abuse and cost	AC-1 to AC-6	6	0
Reliability	RE-1 to RE-4	3	1

5 Out of scope

- Placing the order with the shop and tracking delivery.
- Accounts and sign-in; sharing a wallet between devices or people.
- Currencies other than US dollars; languages other than English.
- Live (non-sandbox) PayPal payments.
- Several purchases from one request, such as a full grocery list.